Yuliang Fan

Curriculum Vitae

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Research Interests: AI for Protein, Deep Generative Models, Multimodal Learning, Reinforcement Learning

Education

2020–2023 Shanghai,CN — **Shanghai Jiao Tong University**, *M.Eng. of EECS*.
GPA:3.5/4.0, Specialized in Computer Vision and Image Processing
Thesis Topic: Deep Learning-Based Reference-Guided Image Inpainting.
Supervisor: Prof. Yue Zhou, Institute of Image Processing and Pattern Recognition.
Awards: JAPAN SCSK Academic Scholarship (Top 5%).

2015–2019 Xi'an,CN — **Xi'an Jiaotong University**, *B.Eng. of Automation Science and Engineering*.
Awards: **1st Prize** in China Undergraduate Mathematical Contest in Modeling, Project Leader in National Undergraduate Innovation and Entrepreneurship Training Program, Outstanding Student in Campus Activities.

Research Experience

AI for Protein

2025–present Shanghai,CN — **Shanghai Jiao Tong University School of Medicine**, *Research Intern*.
Advisor: Haicang Zhang, Professor, Department of Pharmacology and Artificial Intelligence
Project: PocketX: Preference Alignment for Protein Pockets Design through Group Relative Policy Optimization
Developed PocketX, an online reinforcement learning framework for protein pocket design that explicitly aligns a generative model with desired biophysical properties, empowering post-training model to generate protein with high binding affinity and evolutionary plausibility.

Detailed achievements:
- Proposed a hybrid continuous–discrete diffusion framework for pocket structure and sequence co-design, leveraging an AlphaFold3-style score network and online RL (GRPO) with biophysical and evolutionary rewards.
- Extended RL algorithm GRPO, originally developed for the modern autoregressive LLMs (e.g., DeepSeek), to protein pocket design, and demonstrated clear advantages over DPO in terms of reward alignment and training stability through empirical evaluation.
- Extensive experiments show that PocketX significantly outperforms the state-of-the-art protein design model RFdiffusion All-Atom (RFdiffusionAA) in both binding affinity and structural validity on standard benchmarks.

2024–2025 **Westlake University**, *Research Intern*.
Hangzhou,CN Advisor: Fajie Yuan, Assistant Professor, School of Engineering
Project: Toward De Novo Protein Design from Natural Language
Developed Pinal, an innovative deep learning framework for de novo protein design that bridges natural language processing with protein engineering, empowering scientists to generate protein sequences from detailed textual descriptions with precision and flexibility.

Detailed achievements:
- Participated in the design of Pinal's architecture, utilizing a two-stage process for protein sequence design: first translating natural language inputs into structure tokens, then generating sequences based on these structures and text descriptions.
- Contributed to training Pinal with 16 billion parameters on 1.7 billion protein-text pairs, achieving superior performance compared to state-of-the-art methods (e.g., ESM-3).
- Compared with ESM-3, Pinal generates high-quality amino acid sequences with better foldability and functional alignment.

Computer Vision and Image Processing

2020–2023 **Shanghai Jiao Tong University**, *Graduate Researcher*.
Shanghai,CN Advisor: Yue Zhou, Associate Professor, Department of Automation
Project: Multi-Reference Guided Image Inpainting
Detailed achievements:
- Proposed a unified three-stage framework named SLTFill, which explicitly disentangles multi-reference guided image inpainting into multi-reference alignment, spatial transformation, and illumination adjustment.
- Introduced Vision Transformer for reference-guided image inpainting, diving deeper into Transformer for Image Harmonization and Image Alignment as well.
- Independently choose the direction of research topic, proposed experimental design, developed technical method, and published a highly original paper in ICIP 2022.

2019 **Xi'an Jiaotong University**, *Undergraduate Researcher*.
Xi'an,CN
- Image Inpainting Using Generative Adversarial Networks
  - Designed both locally and globally consistent Generative Adversarial Networks to fill arbitrary missing region, resulting in natural image completion.
- Research and Design of Optical Flow-based Quad-Rotor
  - Applied the Harris corner point detector to hover the quad-rotor indoors at the target position.

Publications

2025 **Yuliang Fan**, Haicang Zhang, Jian Zhang, *PocketX: Preference alignment for protein pockets design through group relative policy optimization*, **NeurIPS 2025 Workshop on Machine Learning in Structural Biology (MLSB)**

2025 Jin Su, Zhikai Li, ..., **OPMC Team**, Sergey Ovchinnikov, Fajie Yuan, *Democratizing protein language model training, sharing and collaboration*, **Nature Biotechnology**

2024 Fengyuan Dai, **Yuliang Fan**, Jin Su, Chentong Wang, Chenchen Han, Xibin Zhou, Jianming Liu, Hui Qian, Shunzhi Wang, Anping Zeng, Yajie Wang and Fajie Yuan, *Toward De Novo Protein Design from Natural Language*, bioRxiv 2024.08.01.606258 (**Nature Under Review**)

2022 **Yuliang Fan**, Yue Zhou, Zonghao Yang and Zhenyu Tong, *SLTFILL: Spatial and Light Transformer for Multi-Reference Image Inpainting*, **IEEE International Conference on Image Processing (ICIP)**, Bordeaux, France, 2022

Awards

2022 JAPAN SCSK Academic Scholarship (Top 5%), SJTU
2018 1st Prize in China Undergraduate Mathematical Contest in Modeling, XJTU
2017 National Undergraduate Innovation and Entrepreneurship Training Program, XJTU
2016 Outstanding Student in Campus Activities , XJTU

Leadership Experience

2016–2017 Director of External Relations, Johnson & Johnson Tomorrow Leader Club, XJTU

Services

Fall 2021 Teaching Assistant, Digital Image Processing, SJTU
Fall 2022 Teaching Assistant, Digital Image Processing, SJTU

Skills

Programming Python, Matlab, C++
Packages PyTorch Lighting, PyTorch, TensorFlow, OpenCV, Sklearn
Language Chinese (mother tongue), English (IELTS 7.0)